

✘ Determinants

✘ (Chapter No. 5)

Mathematical Method

5

Consider the simultaneous eqs.

$$a_1x + b_1 = 0 \quad \text{--- (1)}$$

$$a_2x + b_2 = 0 \quad \text{--- (2)}$$

Let us eliminate x from these two eqs.

From (1) $x = -\frac{b_1}{a_1}$

Put in (2)

$$a_2\left(-\frac{b_1}{a_1}\right) + b_2 = 0$$

$$-a_2b_1 + a_1b_2 = 0$$

$$\text{or } a_1b_2 - a_2b_1 = 0$$

The expression on the left i.e., $a_1b_2 - a_2b_1$ is symbolically written as

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = 0 \quad \text{is called a determinant}$$

As this determinant has two rows & two columns, so it is said to be a determinant of order 2. Again

Consider the simultaneous eqs.

$$a_1x + b_1y + c_1 = 0 \quad \text{--- (1)}$$

$$a_2x + b_2y + c_2 = 0 \quad \text{--- (2)}$$

$$a_3x + b_3y + c_3 = 0 \quad \text{--- (3)}$$

Let us eliminate x & y from these three eqs. from (2) & (3)

$$\frac{x}{b_2c_3 - b_3c_2} = \frac{-y}{a_2c_3 - a_3c_2} = \frac{1}{a_2b_3 - a_3b_2}$$

$$\Rightarrow x = \frac{b_2c_3 - b_3c_2}{a_2b_3 - a_3b_2} \quad \text{and} \quad y = -\frac{a_2c_3 - a_3c_2}{a_2b_3 - a_3b_2}$$

Put values in ①

$$a_1 \left(\frac{b_2 c_3 - b_3 c_2}{a_2 b_3 - a_3 b_2} \right) + b_1 \left(- \frac{a_2 c_3 - a_3 c_2}{a_2 b_3 - a_3 b_2} \right) + c_1 = 0$$

$$a_1 (b_2 c_3 - b_3 c_2) - b_1 (a_2 c_3 - a_3 c_2) + c_1 (a_2 b_3 - a_3 b_2) = 0$$

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

As it consists of three rows & three columns, so it is said to be a determinant of order 3.

Properties of determinants:

Following are some important properties of determinants.

- (i) The value of a determinant is the same as the value of its transpose.
- (ii) The interchange of two adjacent rows or columns changes the sign of the determinant.
- (iii) If a row or column of a determinant is passed over m rows or columns then its value is multiplied by $(-1)^m$.
- (iv) If any two rows or columns of a determinant are identical then value of determinant is zero.
- (v) If all the elements in a row or column of a determinant are zero then value of the determinant is zero.
- (vi) If a non zero scalar is multiplied by a determinant then this scalar will be multiplied by any one of the rows or columns of that det.

(vi) If each element in a row or Column of a determinant is the sum of two elements then this determinant will be written as the sum of two determinants as

$$\begin{vmatrix} a_1 & b_1+t_1 & c_1 \\ a_2 & b_2+t_2 & c_2 \\ a_3 & b_3+t_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & t_1 & c_1 \\ a_2 & t_2 & c_2 \\ a_3 & t_3 & c_3 \end{vmatrix}$$

(vii) Addition of some scalar multiple of a row or Column to any other row or Column does not change the value of that determinant.

Minors & Cofactors:

Let $\Delta = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix}$

be a given determinant of order n .

The minor of an element a_{ij} of Δ is the det. M_{ij} obtained by deleting the rows & Columns in which a_{ij} lies. Clearly M_{ij} is a determinant of order $n-1$.

The cofactor A_{ij} of an element a_{ij} of Δ is

$$A_{ij} = (-1)^{i+j} M_{ij}$$

Note

(i) $\Delta = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in}$

for any i

(ii) $\Delta = a_{1j}A_{1j} + a_{2j}A_{2j} + \dots + a_{nj}A_{nj}$

for any j

(iii) If the elements of a line are multiplied by the cofactors of the corresponding elements of any other parallel line & the results so obtained are added the answer will be zero.

Adjoint of a square matrix:

Let $A = [a_{ij}]$ be a square matrix of order n . Denoting the $n \times n$ co-factors by A_{ij} of the elements a_{ij} of A , we define

$$\text{Adj } A = [A_{ij}]_{n \times n}^t = [A_{ji}]_{n \times n}$$

Inverse of a square matrix:

Let A be a non singular square matrix of order n then inverse of A is defined as

$$A^{-1} = \frac{\text{Adj } A}{|A|}$$

Note If A & B are square matrices of order n then

- (i) $\det(AB) = \det(A) \cdot \det(B)$
- (ii) $\det(BA) = \det(B) \cdot \det(A)$
- (iii) $\det(A^{-1}) = (\det(A))^{-1}$ if A is non singular
- (iv) $\det(A^t) = \det(A)$
- (v) $\det(A^n) = [\det(A)]^n$ where $n \in \mathbb{Z}^+$
- (vi) $\det(KA) = K^n \cdot \det(A)$

✿ Exercise No. 5.1 ✿

Q1 Let M_2 be the set of all 2×2 matrices.

Set up the transformation $A \rightarrow \det(A)$, $A \in M_2$.

What is the range of this mapping?

Is the mapping one-to-one?

Sol.

5 Mathematical Method

Let $f: A \rightarrow \det(A)$; $A \in M_2$
be defined by

$$f(A) = \det(A)$$

Suppose the field for all $A \in M_2$ be the set of Complex no's C , then the range of f is C . But if the field is taken as the set of real no's R then range of f is also R .

This mapping f is not one-to-one as shown by the following example

$$\text{Let } A = \begin{bmatrix} 2 & 2 \\ 1 & 4 \end{bmatrix} + B = \begin{bmatrix} 6 & 0 \\ 9 & 1 \end{bmatrix}$$

then clearly $A \neq B$

Now

$$\det(A) = \begin{vmatrix} 2 & 2 \\ 1 & 4 \end{vmatrix} = 8 - 2 = 6$$

$$+ \det(B) = \begin{vmatrix} 6 & 0 \\ 9 & 1 \end{vmatrix} = 6 - 0 = 6$$

Hence $\det(A) = \det(B)$

So we have proved that

$$A \neq B \Rightarrow \det(A) = \det(B)$$

Hence by def., f is not one-to-one.

Q2 For 2×2 matrices A & B which of the following equations hold?

(i) $\det(A+B) = \det(A) + \det(B)$

Sol.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} f & g \\ h & k \end{bmatrix}$

then

$$A+B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} f & g \\ h & k \end{bmatrix}$$

or

$$A+B = \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix}$$

$$\Rightarrow \det(A+B) = \begin{vmatrix} a+f & b+g \\ c+h & d+k \end{vmatrix}$$

$$= (a+f)(d+k) - (b+g)(c+h)$$

$$= ad+ak+fd+fk - bc-bh-gc-gh$$

①

Now

$$\det A + \det B = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} f & g \\ h & k \end{vmatrix}$$

$$= ad-bc+fk-gh$$

②

from ① & ②

$$\det(A+B) \neq \det A + \det B$$

(ii) $\det(A+B)^2 = [\det(A+B)]^2$

Sol.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} f & g \\ h & k \end{bmatrix}$

then

$$A+B = \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} f & g \\ h & k \end{bmatrix}$$

$$\text{or } A+B = \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix}$$

427

Now

$$(A+B)^2 = \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix} \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix}$$

$$\therefore (A+B)^2 = \begin{bmatrix} (a+f)^2 + (b+g)(c+h) & (a+f)(b+g) + (b+g)(d+k) \\ (c+h)(a+f) + (d+k)(c+h) & (c+h)(b+g) + (d+k)^2 \end{bmatrix}$$

So

$$\det(A+B)^2 = \begin{vmatrix} (a+f)^2 + (b+g)(c+h) & (a+f)(b+g) + (b+g)(d+k) \\ (c+h)(a+f) + (d+k)(c+h) & (c+h)(b+g) + (d+k)^2 \end{vmatrix} \quad \text{--- (1)}$$

Now

$$A+B = \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix}$$

then

$$\det(A+B) = \begin{vmatrix} a+f & b+g \\ c+h & d+k \end{vmatrix}$$

$$\therefore [\det(A+B)]^2 = \begin{vmatrix} a+f & b+g \\ c+h & d+k \end{vmatrix} \begin{vmatrix} a+f & b+g \\ c+h & d+k \end{vmatrix}$$

$$\therefore [\det(A+B)]^2 = \begin{vmatrix} (a+f)^2 + (b+g)(c+h) & (a+f)(b+g) + (b+g)(d+k) \\ (c+h)(a+f) + (d+k)(c+h) & (c+h)(b+g) + (d+k)^2 \end{vmatrix} \quad \text{--- (2)}$$

from (1) & (2)

$$\det(A+B)^2 = [\det(A+B)]^2$$

$$(iii) \quad \det(A+B)^2 = \det(A^2+B^2)$$

Sol. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ & $B = \begin{bmatrix} f & g \\ h & k \end{bmatrix}$

then

$$A+B = \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix}$$

Now

$$(A+B)^2 = \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix} \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix}$$

$$(A+B)^2 = \begin{bmatrix} (a+f)^2 + (b+g)(c+h) & (a+f)(b+g) + (b+g)(d+k) \\ (c+h)(a+f) + (d+k)(c+h) & (c+h)(b+g) + (d+k)^2 \end{bmatrix}$$

So

$$\det(A+B)^2 = \begin{vmatrix} (a+f)^2 + (b+g)(c+h) & (a+f)(b+g) + (b+g)(d+k) \\ (c+h)(a+f) + (d+k)(c+h) & (c+h)(b+g) + (d+k)^2 \end{vmatrix} \quad \text{--- ①}$$

Now

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$= \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} f & g \\ h & k \end{bmatrix} \begin{bmatrix} f & g \\ h & k \end{bmatrix}$$

$$= \begin{bmatrix} f^2+gh & fg+gk \\ hf+kh & gh+k^2 \end{bmatrix}$$

So

$$A^2+B^2 = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix} + \begin{bmatrix} f^2+gh & fg+gk \\ hf+kh & gh+k^2 \end{bmatrix}$$

$$A^2+B^2 = \begin{bmatrix} a^2+f^2+bc+gh & ab+bd+fg+gk \\ ac+cd+hf+kh & d^2+k^2+bc+gh \end{bmatrix}$$

So

$$\det(A^2+B^2) = \begin{vmatrix} a^2+f^2+bc+gh & ab+bd+fg+gk \\ ac+cd+hf+kh & d^2+k^2+bc+gh \end{vmatrix} \quad \text{--- ②}$$

from ① + ②

$$\det(A+B)^2 \neq \det(A^2+B^2)$$

$$(iv) \det(A+B)^2 = \det(A^2+2AB+B^2)$$

Soln.

$$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } B = \begin{bmatrix} f & g \\ h & k \end{bmatrix}$$

Then

$$A+B = \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix}$$

Now

$$(A+B)^2 = \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix} \begin{bmatrix} a+f & b+g \\ c+h & d+k \end{bmatrix}$$

$$(A+B)^2 = \begin{bmatrix} (a+f)^2 + (b+g)(c+h) & (a+f)(b+g) + (b+g)(d+k) \\ (c+h)(a+f) + (d+k)(c+h) & (c+h)(b+g) + (d+k)^2 \end{bmatrix}$$

So

$$\det(A+B)^2 = \begin{vmatrix} (a+f)^2 + (b+g)(c+h) & (a+f)(b+g) + (b+g)(d+k) \\ (c+h)(a+f) + (d+k)(c+h) & (c+h)(b+g) + (d+k)^2 \end{vmatrix} \quad \text{--- (1)}$$

Now

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} f & g \\ h & k \end{bmatrix} \begin{bmatrix} f & g \\ h & k \end{bmatrix} = \begin{bmatrix} f^2+gh & fg+gk \\ hf+kh & gh+k^2 \end{bmatrix}$$

4

$$2AB = 2 \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} f & g \\ h & k \end{bmatrix}$$

$$= 2 \begin{bmatrix} af+bh & ag+bk \\ cf+dh & gc+dk \end{bmatrix}$$

$$= \begin{bmatrix} 2af+2bh & 2ag+2bk \\ 2cf+2dh & 2gc+2dk \end{bmatrix}$$

So

$$A^2 + 2AB + B^2 = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix} + \begin{bmatrix} 2af+2bh & 2ag+2bk \\ 2cf+2dh & 2gc+2dk \end{bmatrix} + \begin{bmatrix} f^2+gh & fg+gk \\ hf+kh & gh+k^2 \end{bmatrix}$$

$$A^2 + 2AB + B^2 = \begin{bmatrix} a^2+bc+2af+2bh+f^2+gh & ab+bd+2ag+2bk+fg+gk \\ ac+cd+2cf+2dh+hf+kh & bc+d^2+2gc+2dk+gh+k^2 \end{bmatrix}$$

So

$$\det(A^2 + 2AB + B^2) = \begin{vmatrix} a^2+bc+2af+2bh+f^2+gh & ab+bd+2ag+2bk+fg+gk \\ ac+cd+2cf+2dh+hf+kh & bc+d^2+2gc+2dk+gh+k^2 \end{vmatrix} \quad \text{--- (2)}$$

from (1) & (2)

$$\det(A+B)^2 \neq \det(A^2 + 2AB + B^2)$$

Q13 Find the value of each of the following determinants:

(1) $\begin{vmatrix} 1 & 0 & 2 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{vmatrix}$ PUACP

Soln

$$\text{Let } \Delta = \begin{vmatrix} 1 & 0 & 2 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{vmatrix}$$

Expanding from R_1

$$= 1 \begin{vmatrix} 4 & 5 \\ 6 & 7 \end{vmatrix} - 0 \begin{vmatrix} 3 & 5 \\ 5 & 7 \end{vmatrix} + 2 \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix}$$

$$= 1(28-30) - 0 + 2(18-20)$$

$$= -2 + 2(-2)$$

$$= -2 - 4$$

$$\Delta = -6$$

$$(ii) \begin{vmatrix} 2 & -1 & 1 \\ 3 & 2 & 4 \\ -1 & 0 & 3 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 2 & -1 & 1 \\ 3 & 2 & 4 \\ -1 & 0 & 3 \end{vmatrix}$$

Expanding from R_1

$$= 2 \begin{vmatrix} 2 & 4 \\ 0 & 3 \end{vmatrix} + 1 \begin{vmatrix} 3 & 4 \\ -1 & 3 \end{vmatrix} + 1 \begin{vmatrix} 3 & 2 \\ -1 & 0 \end{vmatrix}$$

$$= 2(6-0) + (9+4) + (0+2)$$

$$= 2(6) + 13 + 2$$

$$= 12 + 15$$

$$\Delta = 27$$

(iii)

$$\begin{vmatrix} 6 & -6 & 6 \\ 2 & 4 & -6 \\ 15 & -5 & 5 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 6 & -6 & 6 \\ 2 & 4 & -6 \\ 15 & -5 & 5 \end{vmatrix}$$

Expanding from R_1

$$= 6 \begin{vmatrix} 4 & -6 \\ -5 & 5 \end{vmatrix} + 6 \begin{vmatrix} 2 & -6 \\ 15 & 5 \end{vmatrix} + 6 \begin{vmatrix} 2 & 4 \\ 15 & -5 \end{vmatrix}$$

$$= 6(20-30) + 6(10+90) + 6(-10-60)$$

$$= 6(-10) + 6(100) + 6(-70)$$

$$= -60 + 600 - 420$$

$$\Delta = -60 + 180 = 120$$

Q4 Evaluate

$$(i) \begin{vmatrix} 2 & 3 & -2 & 4 \\ 7 & 4 & -3 & 10 \\ 3 & 2 & 3 & 4 \\ -2 & 4 & 0 & 5 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 2 & 3 & -2 & 4 \\ 7 & 4 & -3 & 10 \\ 3 & 2 & 3 & 4 \\ -2 & 4 & 0 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 3 & -2 & 4 \\ 7 & 4 & -3 & 10 \\ 1 & -1 & 5 & 0 \\ -2 & 4 & 0 & 5 \end{vmatrix} \quad R_3 - R_1$$

$$= \begin{vmatrix} 0 & 5 & -12 & 4 \\ 0 & 11 & -38 & 10 \\ 1 & -1 & 5 & 0 \\ 0 & 2 & 10 & 5 \end{vmatrix} \quad \begin{array}{l} R_1 - 2R_3 \\ R_2 - 7R_3 \\ R_4 + 2R_3 \end{array}$$

Expanding from C_1

$$= 0 - 0 + \begin{vmatrix} 5 & -12 & 4 \\ 11 & -38 & 10 \\ 2 & 10 & 5 \end{vmatrix} - 0$$

$$= \begin{vmatrix} 5 & -12 & 4 \\ 11 & -38 & 10 \\ 2 & 10 & 5 \end{vmatrix}$$

Expanding from R_1

$$= 5 \begin{vmatrix} -38 & 10 \\ 10 & 5 \end{vmatrix} + 12 \begin{vmatrix} 11 & 10 \\ 2 & 5 \end{vmatrix} + 4 \begin{vmatrix} 11 & -38 \\ 2 & 10 \end{vmatrix}$$

$$= 5(-190 - 100) + 12(55 - 20) + 4(110 + 76)$$

$$= 5(-290) + 12(35) + 4(186)$$

$$= -1450 + 420 + 744$$

$$= -1450 + 1164$$

$$\Delta = -286$$

(ii)

$$\begin{vmatrix} 3 & 7 & 5 & 2 \\ 2 & 4 & 1 & 1 \\ -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 4 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 3 & 7 & 5 & 2 \\ 2 & 4 & 1 & 1 \\ -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 4 \end{vmatrix}$$

Expanding from R_3

$$= -2 \begin{vmatrix} 7 & 5 & 2 \\ 4 & 1 & 1 \\ 1 & 3 & 4 \end{vmatrix}$$

Expanding from R_1

$$= -2 \left\{ 7 \begin{vmatrix} 1 & 1 \\ 3 & 4 \end{vmatrix} - 5 \begin{vmatrix} 4 & 1 \\ 1 & 4 \end{vmatrix} + 2 \begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} \right\}$$

$$= -2 \left\{ 7(4-3) - 5(16-1) + 2(12-1) \right\}$$

$$= -2 \left\{ 7(1) - 5(15) + 2(11) \right\}$$

$$= -2(7-75+22)$$

$$= -2(-46)$$

$$= 92$$

(iii)
$$\begin{vmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 1 & 3 & 0 & -3 \\ 0 & -7 & 3 & 1 \end{vmatrix}$$

Sol.

Let $\Delta = \begin{vmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 1 & 3 & 0 & -3 \\ 0 & -7 & 3 & 1 \end{vmatrix}$

$$= \begin{vmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 0 & 5 & -3 & 1 \\ 0 & -7 & 3 & 1 \end{vmatrix}$$

$R_3 - R_1$

$$= \begin{vmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ -7 & 3 & 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 0 & -4 & 8 & 1 \end{vmatrix}$$

$R_2 - 5R_1$

$R_3 + 7R_1$

Expanding from C_1

$$= 1 \begin{vmatrix} 2 & -4 \\ -4 & 8 \end{vmatrix}$$

$$= 16 - 16$$

$$\Delta = 0$$

(iv)

$$\begin{vmatrix} 9 & 93 & 12 & -6 \\ 1 & 92 & 84 & -6 \\ 2 & 185 & 100 & -12 \\ 4 & 270 & 196 & -24 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 9 & 93 & 12 & -6 \\ 1 & 92 & 84 & -6 \\ 2 & 185 & 100 & -12 \\ 4 & 270 & 196 & -24 \end{vmatrix}$$

taking -6 Common from C_4

$$= -6 \begin{vmatrix} 9 & 93 & 12 & 1 \\ 1 & 92 & 84 & 1 \\ 2 & 185 & 100 & 2 \\ 4 & 270 & 196 & 4 \end{vmatrix}$$

$$= -6 \begin{vmatrix} 9 & 93 & 12 & 1 \\ -8 & -1 & 72 & 0 \\ -16 & -1 & 76 & 0 \\ -32 & -1.2 & 148 & 0 \end{vmatrix}$$

$$R_2 - R_1$$

$$R_3 - 2R_1$$

$$R_4 - 4R_1$$

Expanding from C_4

$$= -6 \begin{vmatrix} -8 & -1 & 72 \\ -16 & -1 & 76 \\ -32 & -1.2 & 148 \end{vmatrix}$$

taking $-8, -1, 4$ Common from C_1, C_2, C_3

$$= (-6)(-8)(-1)(4) \begin{vmatrix} 1 & 1 & 18 \\ 2 & 1 & 19 \\ 4 & 1.2 & 37 \end{vmatrix}$$

$$= -192 \begin{vmatrix} 1 & 1 & 18 \\ 2 & 1 & 19 \\ 4 & 1.2 & 37 \end{vmatrix}$$

$$= -192 \begin{vmatrix} 1 & 1 & 18 \\ 0 & -1 & -17 \\ 0 & 98 & -35 \end{vmatrix}$$

$$R_2 - 2R_1$$

$$R_3 - 4R_1$$

Expanding from C_1

$$= -192 \begin{vmatrix} -1 & -17 \\ 98 & -35 \end{vmatrix}$$

$$\Delta = -192(35 + 1666)$$

$$= -192(1701)$$

$$\Delta = -326592$$

(v)
$$\begin{vmatrix} 1 & 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 & 1 \\ -1 & -1 & -1 & 1 & 1 \end{vmatrix}$$

Sol.

Let
$$\Delta = \begin{vmatrix} 1 & 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 & 1 \\ -1 & -1 & -1 & 1 & 1 \end{vmatrix}$$

$$\times \begin{vmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & -2 & 2 & 0 & 0 \\ -1 & 2 & -2 & 0 & -2 \\ 1 & -2 & 0 & 0 & 2 \\ -1 & 0 & -2 & 0 & 0 \end{vmatrix}$$

$C_2 - C_1$
 $C_3 + C_1$
 $C_4 + C_1$
 $C_5 + C_1$

$$= 0 \quad (\because C_4 = 0)$$

So $\Delta = 0$

Q8 Without expanding, show that

(i)
$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & k \end{vmatrix} = \begin{vmatrix} e & b & h \\ d & a & g \\ f & c & k \end{vmatrix}$$

Sol.

Sol.

Consider $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & k \end{vmatrix}$

$$= \begin{vmatrix} d & e & f \\ a & b & c \\ g & h & k \end{vmatrix}$$

 R_{12}

$$= \begin{vmatrix} e & d & f \\ b & a & c \\ h & g & k \end{vmatrix}$$

 C_{12}

$$= \begin{vmatrix} e & b & h \\ d & a & g \\ f & c & k \end{vmatrix}$$

By taking transpose

So

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & k \end{vmatrix} = \begin{vmatrix} e & b & h \\ d & a & g \\ f & c & k \end{vmatrix}$$

(ii)

$$\begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix} = 0$$

Sol.

Let $\Delta = \begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix}$

$$= \begin{vmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{vmatrix}$$

By taking transpose

$$= (-1)^3 \begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix}$$

By taking -1 common from R_1, R_2, R_3

$$\Delta = -\Delta$$

$$\Delta + \Delta = 0$$

$$2\Delta = 0$$

$$\Delta = 0$$

$$\begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix} = 0$$

$$(iii) \begin{vmatrix} a+b & c & 1 \\ b+c & a & 1 \\ c+a & b & 1 \end{vmatrix} = 0$$

Sol.

$$\text{let } \Delta = \begin{vmatrix} a+b & c & 1 \\ b+c & a & 1 \\ c+a & b & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a+b+c & c & 1 \\ b+c+a & a & 1 \\ c+a+b & b & 1 \end{vmatrix} \quad C_1 + C_2$$

$$= \begin{vmatrix} a+b+c & c & 1 \\ a+b+c & a & 1 \\ a+b+c & b & 1 \end{vmatrix}$$

$$= (a+b+c) \begin{vmatrix} 1 & c & 1 \\ 1 & a & 1 \\ 1 & b & 1 \end{vmatrix}$$

taking $a+b+c$ common from C_1

$$= (a+b+c)(0)$$

$$\therefore C_1 = C_3$$

$$\Delta = 0$$

$$\text{or } \begin{vmatrix} a+b & c & 1 \\ b+c & a & 1 \\ c+a & b & 1 \end{vmatrix} = 0$$

Q6 Prove that

$$(i) \begin{vmatrix} bc & ca & ab \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ a^2 & b^2 & c^2 \end{vmatrix} = 0$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} bc & ca & ab \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} bc & ca & ab \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix}$$

Multiplying R_2 by abc

$$= \frac{1}{abc} (0) \quad \therefore R_1 = R_2$$

$$\Delta = 0$$

So $\begin{vmatrix} bc & ca & ab \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ a^2 & b^2 & c^2 \end{vmatrix} = 0$

$$(ii) \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

Available at
www.mathcity.org

$$= \begin{vmatrix} a-b+b-c+c-a & b-c & c-a \\ b-c+c-a+a-b & c-a & a-b \\ c-a+a-b+b-c & a-b & b-c \end{vmatrix} \quad C_1 + (C_2 + C_3)$$

$$= \begin{vmatrix} 0 & b-c & c-a \\ 0 & c-a & a-b \\ 0 & a-b & b-c \end{vmatrix}$$

$$\Delta = 0 \quad \therefore C_1 = 0$$

$$S_0 \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$$

$$(iii) \begin{vmatrix} a & a^2 & a/bc \\ b & b^2 & b/ca \\ c & c^2 & c/ab \end{vmatrix} = 0$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} a & a^2 & a/bc \\ b & b^2 & b/ca \\ c & c^2 & c/ab \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} a & a^2 & a^2 \\ b & b^2 & b^2 \\ c & c^2 & c^2 \end{vmatrix}$$

Multiplying C_3 by abc

$$\Delta = \frac{1}{abc} (0)$$

$$\Delta = 0$$

So

$$\begin{vmatrix} a & a^2 & a/bc \\ b & b^2 & b/ca \\ c & c^2 & c/ab \end{vmatrix} = 0$$

(iv)

$$\begin{vmatrix} \sin^2 \theta & 1 & \cos^2 \theta \\ \sin^2 \phi & 1 & \cos^2 \phi \\ \sin^2 \psi & 1 & \cos^2 \psi \end{vmatrix} = 0$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} \sin^2 \theta & 1 & \cos^2 \theta \\ \sin^2 \phi & 1 & \cos^2 \phi \\ \sin^2 \psi & 1 & \cos^2 \psi \end{vmatrix}$$

$$= \begin{vmatrix} \sin^2 \theta + \cos^2 \theta & 1 & \cos^2 \theta \\ \sin^2 \phi + \cos^2 \phi & 1 & \cos^2 \phi \\ \sin^2 \psi + \cos^2 \psi & 1 & \cos^2 \psi \end{vmatrix} \quad R_1 + R_2$$

$$= \begin{vmatrix} 1 & 1 & \cos^2 \theta \\ 1 & 1 & \cos^2 \phi \\ 1 & 1 & \cos^2 \psi \end{vmatrix}$$

$$\Delta = 0$$

$$\therefore C_1 = C_2$$

$$\begin{vmatrix} \sin^2 \theta & 1 & \cos^2 \theta \\ \sin^2 \phi & 1 & \cos^2 \phi \\ \sin^2 \psi & 1 & \cos^2 \psi \end{vmatrix} = 0$$



(v)

$$\begin{vmatrix} \sin^2 \alpha & \cos^2 \alpha & \cos^2 \alpha \\ \sin^2 \beta & \cos^2 \beta & \cos^2 \beta \\ \sin^2 \gamma & \cos^2 \gamma & \cos^2 \gamma \end{vmatrix} = 0$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} \sin^2 d & \cos^2 d & \cos^2 d \\ \sin^2 p & \cos^2 p & \cos^2 p \\ \sin^2 y & \cos^2 y & \cos^2 y \end{vmatrix}$$

$$= \begin{vmatrix} \sin^2 d & \cos^2 d & \cos^2 d - \sin^2 d \\ \sin^2 p & \cos^2 p & \cos^2 p - \sin^2 p \\ \sin^2 y & \cos^2 y & \cos^2 y - \sin^2 y \end{vmatrix} \quad C_3 - C_1$$

$$= \begin{vmatrix} \sin^2 d & \cos^2 d & \cos^2 d \\ \sin^2 p & \cos^2 p & \cos^2 p \\ \sin^2 y & \cos^2 y & \cos^2 y \end{vmatrix} \quad \therefore \cos^2 \theta = \cos^2 \theta - \sin^2 \theta$$

$$\Delta = 0 \quad \therefore C_1 = C_3$$

So,

$$\begin{vmatrix} \sin^2 d & \cos^2 d & \cos^2 d \\ \sin^2 p & \cos^2 p & \cos^2 p \\ \sin^2 y & \cos^2 y & \cos^2 y \end{vmatrix} = 0$$

(vi)

$$\begin{vmatrix} \cos d & \sin d & \sin(d+\delta) \\ \cos p & \sin p & \sin(p+\delta) \\ \cos y & \sin y & \sin(y+\delta) \end{vmatrix} = 0$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} \cos d & \sin d & \sin(d+\delta) \\ \cos p & \sin p & \sin(p+\delta) \\ \cos y & \sin y & \sin(y+\delta) \end{vmatrix}$$

$$\Delta = \begin{vmatrix} \cos d & \sin d & \sin d \cos \delta + \cos d \sin \delta \\ \cos p & \sin p & \sin p \cos \delta + \cos p \sin \delta \\ \cos y & \sin y & \sin y \cos \delta + \cos y \sin \delta \end{vmatrix}$$

$$= \begin{vmatrix} \cos d & \sin d & \sin d \cos \delta \\ \cos p & \sin p & \sin p \cos \delta \\ \cos y & \sin y & \sin y \cos \delta \end{vmatrix} + \begin{vmatrix} \cos d & \sin d & \cos d \sin \delta \\ \cos p & \sin p & \cos p \sin \delta \\ \cos y & \sin y & \cos y \sin \delta \end{vmatrix}$$

$$= \cos \delta \begin{vmatrix} \cos d & \sin d & \sin d \\ \cos p & \sin p & \sin p \\ \cos y & \sin y & \sin y \end{vmatrix} + \sin \delta \begin{vmatrix} \cos d & \sin d & \cos d \\ \cos p & \sin p & \cos p \\ \cos y & \sin y & \cos y \end{vmatrix}$$

$$= \cos \delta (0) + \sin \delta (0) \quad (\because \text{two columns are identical})$$

$$= 0 + 0$$

$$\Delta = 0$$

$$\text{So } \begin{vmatrix} \cos d & \sin d & \sin(d+\delta) \\ \cos p & \sin p & \sin(p+\delta) \\ \cos y & \sin y & \sin(y+\delta) \end{vmatrix} = 0$$

(vii)

$$\begin{vmatrix} 1 & \cos d & \cos p \\ \cos d & 1 & \cos(d+p) \\ \cos p & \cos(d+p) & 1 \end{vmatrix} = 0$$

Sol.

$$\Delta = \begin{vmatrix} 1 & \cos d & \cos p \\ \cos d & 1 & \cos d \cos p - \sin d \sin p \\ \cos p & \cos d \cos p - \sin d \sin p & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ \cos d & 1 - \cos^2 d & -\sin d \sin p \\ \cos p & -\sin d \sin p & 1 - \cos^2 p \end{vmatrix}$$

$$C_2 - (\cos d)C_1$$

$$C_3 - (\cos p)C_1$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ \cos d & \sin^2 d & -\sin d \sin p \\ \cos p & -\sin d \sin p & \sin^2 p \end{vmatrix}$$

$$= (\sin d)(-\sin p) \begin{vmatrix} 1 & 0 & 0 \\ \cos d & \sin d & \sin d \\ \cos p & -\sin p & -\sin p \end{vmatrix} \quad \begin{array}{l} \text{taking } \sin d \text{ common from } C_2 \\ d - \sin p \text{ common from } C_3 \end{array}$$

$$= -\sin d \sin p (0)$$

$$\therefore C_2 = C_3$$

$$\Delta = 0$$

$$\text{So } \begin{vmatrix} 1 & \cos d & \cos p \\ \cos d & 1 & \cos(d+p) \\ \cos p & \cos(d+p) & 1 \end{vmatrix} = 0$$

$$(viii) \begin{vmatrix} (a+b)^2 & a^2+b^2 & ab \\ (c+d)^2 & c^2+d^2 & cd \\ (g+h)^2 & g^2+h^2 & gh \end{vmatrix} = 0$$



Sol.

$$\text{Let } \Delta = \begin{vmatrix} (a+b)^2 & a^2+b^2 & ab \\ (c+d)^2 & c^2+d^2 & cd \\ (g+h)^2 & g^2+h^2 & gh \end{vmatrix}$$

$$= \begin{vmatrix} a^2+b^2+2ab & a^2+b^2 & ab \\ c^2+d^2+2cd & c^2+d^2 & cd \\ g^2+h^2+2gh & g^2+h^2 & gh \end{vmatrix}$$

$$= \begin{vmatrix} 2ab & a^2+b^2 & ab \\ 2cd & c^2+d^2 & cd \\ 2gh & g^2+h^2 & gh \end{vmatrix} \quad C_1 - C_2$$

$$= 2 \begin{vmatrix} ab & a^2+b^2 & ab \\ cd & c^2+d^2 & cd \\ gh & g^2+h^2 & gh \end{vmatrix} \quad \text{taking 2 Common from } C_1$$

$$= 2(0) \quad (\because C_1 = C_3)$$

$$\Delta = 0$$

$$S. \quad \begin{vmatrix} (a+b)^2 & a^2+b^2 & ab \\ (c+d)^2 & c^2+d^2 & cd \\ (g+h)^2 & g^2+h^2 & gh \end{vmatrix} = 0$$

$$(ix) \begin{vmatrix} (a^m + \bar{a}^m)^2 & (a^m - \bar{a}^m)^2 & abc \\ (b^n + \bar{b}^n)^2 & (b^n - \bar{b}^n)^2 & abc \\ (c^p + \bar{c}^p)^2 & (c^p - \bar{c}^p)^2 & abc \end{vmatrix} = 0$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} (a^m + \bar{a}^m)^2 & (a^m - \bar{a}^m)^2 & abc \\ (b^n + \bar{b}^n)^2 & (b^n - \bar{b}^n)^2 & abc \\ (c^p + \bar{c}^p)^2 & (c^p - \bar{c}^p)^2 & abc \end{vmatrix}$$

$$= abc \begin{vmatrix} \frac{2m}{a+a} + 2 & \frac{2m}{a+a} - 2 & 1 \\ \frac{2n}{b+b} + 2 & \frac{2n}{b+b} - 2 & 1 \\ \frac{2p}{c+c} + 2 & \frac{2p}{c+c} - 2 & 1 \end{vmatrix}$$

$$= abc \begin{vmatrix} \frac{2m}{a+a} - \frac{2m}{a+a} & \frac{2m}{a+a} - \frac{2m}{a+a} & 1 \\ \frac{2n}{b+b} - \frac{2n}{b+b} & \frac{2n}{b+b} - \frac{2n}{b+b} & 1 \\ \frac{2p}{c+c} - \frac{2p}{c+c} & \frac{2p}{c+c} - \frac{2p}{c+c} & 1 \end{vmatrix}$$

$$C_1 - 2C_3$$

$$C_2 + 2C_3$$

$$= (abc)(0)$$

$$\therefore C_1 = C_2$$

$$\Delta = 0$$

$$(x) \begin{vmatrix} \frac{1}{2!} & 1 & 0 \\ \frac{1}{3!} & \frac{1}{2!} & 1 \\ \frac{1}{4!} & \frac{1}{3!} & \frac{1}{2!} \end{vmatrix} = 0$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} \frac{1}{2!} & 1 & 0 \\ \frac{1}{3!} & \frac{1}{2!} & 1 \\ \frac{1}{4!} & \frac{1}{3!} & \frac{1}{2!} \end{vmatrix}$$

$$\Delta = \begin{vmatrix} \frac{1}{2} & 1 & 0 \\ \frac{1}{6} & \frac{1}{2} & 1 \\ \frac{1}{24} & \frac{1}{6} & \frac{1}{2} \end{vmatrix}$$

$$= \left(\frac{1}{2}\right)\left(\frac{1}{6}\right)\left(\frac{1}{24}\right) \begin{vmatrix} 1 & 2 & 0 \\ 1 & 3 & 6 \\ 1 & 4 & 12 \end{vmatrix}$$

taking $\frac{1}{2}, \frac{1}{6}, \frac{1}{24}$ common from R_1, R_2, R_3

$$= \frac{1}{288} \begin{vmatrix} 1 & 2 & 0 \\ 1 & 3 & 6 \\ 1 & 4 & 12 \end{vmatrix}$$

$$= \frac{1}{288} \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 6 \\ 1 & 2 & 12 \end{vmatrix}$$

$$= \frac{6}{288} \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 2 \end{vmatrix}$$

taking 6 common from C_3

$$= \frac{1}{48} (0)$$

$C_2 = C_3$

$$\Delta = 0$$

(xi)

$$\begin{vmatrix} a^2 & b \sin d & c \sin d \\ b \sin d & 1 & c \sin d \\ c \sin d & c \sin d & 1 \end{vmatrix} = 0$$

where a, b, c are the magnitudes of the sides of a triangle & d is the measure of the angle opposite to the side with magnitude a .

$$= \begin{vmatrix} a^2 - c^2 \sin^2 d & b \sin d - c \sin d \cos d & 0 \\ b \sin d - c \sin d \cos d & 1 - \cos^2 d & 0 \\ c \sin d & \cos d & 1 \end{vmatrix} \quad \begin{array}{l} R_1 - c \sin d R_3 \\ R_2 - \cos d R_3 \end{array}$$

Expanding from C_3

$$= \begin{vmatrix} a^2 - c^2 \sin^2 d & \sin d (b - c \cos d) \\ \sin d (b - c \cos d) & \sin^2 d \end{vmatrix}$$

$$\begin{aligned} &= \sin^2 d (a^2 - c^2 \sin^2 d) - \sin^2 d (b - c \cos d)^2 \\ &= a^2 \sin^2 d - c^2 \sin^4 d - \sin^2 d (b^2 + c^2 \cos^2 d - 2bc \cos d) \\ &= a^2 \sin^2 d - c^2 \sin^4 d - b^2 \sin^2 d - c^2 \sin^2 d \cos^2 d + 2bc \sin^2 d \cos d \\ &= a^2 \sin^2 d - c^2 \sin^4 d - b^2 \sin^2 d - c^2 \sin^2 d (1 - \sin^2 d) + 2bc \sin^2 d \cos d \\ &= a^2 \sin^2 d - c^2 \sin^4 d - b^2 \sin^2 d - c^2 \sin^2 d + c^2 \sin^4 d + 2bc \sin^2 d \cos d \\ &= [a^2 - b^2 - c^2 + 2bc \cos d] \sin^2 d \\ &= \left[a^2 - b^2 - c^2 + 2bc \left(\frac{b^2 + c^2 - a^2}{2bc} \right) \right] \sin^2 d \\ &= [a^2 - b^2 - c^2 + b^2 + c^2 - a^2] \sin^2 d \\ &= (0) \sin^2 d \end{aligned}$$

$$\Delta = 0$$

(xii)

$$\begin{vmatrix} a & b & c & d & 1 \\ b & c & d & a & 1 \\ c & d & a & b & 1 \\ d & a & b & c & 1 \\ b & a & d & c & 1 \end{vmatrix} = 0$$

Sol:-

$$\text{Let } \Delta = \begin{vmatrix} a & b & c & d & 1 \\ b & c & d & a & 1 \\ c & d & a & b & 1 \\ d & a & b & c & 1 \\ b & a & d & c & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a+b+c+d & b & c & d & 1 \\ b+c+d+a & c & d & a & 1 \\ c+d+a+b & d & a & b & 1 \\ d+a+b+c & a & b & c & 1 \\ b+a+d+c & a & d & c & 1 \end{vmatrix}$$

$$C_1 + (C_2 + C_3 + C_4)$$

$$= \begin{vmatrix} a+b+c+d & b & c & d & 1 \\ a+b+c+d & c & d & a & 1 \\ a+b+c+d & d & a & b & 1 \\ a+b+c+d & a & b & c & 1 \\ a+b+c+d & a & d & c & 1 \end{vmatrix}$$

$$= (a+b+c+d) \begin{vmatrix} 1 & b & c & d & 1 \\ 1 & c & d & a & 1 \\ 1 & d & a & b & 1 \\ 1 & a & b & c & 1 \\ 1 & a & d & c & 1 \end{vmatrix}$$

$$= (a+b+c+d)(0)$$

$$\therefore C_1 = C_5$$

$$= 0$$

$$\text{So } \Delta = 0$$

$$(xiii) \begin{vmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & a^2+2a+1 & a^2+4a+4 & a^2+6a+9 \\ b^2 & b^2+2b+1 & b^2+4b+4 & b^2+6b+9 \\ c^2 & c^2+2c+1 & c^2+4c+4 & c^2+6c+9 \\ d^2 & d^2+2d+1 & d^2+4d+4 & d^2+6d+9 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & 2a+1 & 4a+4 & 6a+9 \\ b^2 & 2b+1 & 4b+4 & 6b+9 \\ c^2 & 2c+1 & 4c+4 & 6c+9 \\ d^2 & 2d+1 & 4d+4 & 6d+9 \end{vmatrix} \quad \begin{array}{l} C_2 - C_1 \\ C_3 - C_1 \\ C_4 - C_1 \end{array}$$

$$= \begin{vmatrix} a^2 & 2a+1 & 2 & 6 \\ b^2 & 2b+1 & 2 & 6 \\ c^2 & 2c+1 & 2 & 6 \\ d^2 & 2d+1 & 2 & 6 \end{vmatrix} \quad \begin{array}{l} C_3 - 2C_2 \\ C_4 - 3C_2 \end{array}$$

$$= 3 \begin{vmatrix} a^2 & 2a+1 & 2 & 2 \\ b^2 & 2b+1 & 2 & 2 \\ c^2 & 2c+1 & 2 & 2 \\ d^2 & 2d+1 & 2 & 2 \end{vmatrix}$$

taking 3 Common from C_4

$$\Delta = 0$$

$$\therefore C_3 \neq C_4$$

Q7 Without expansion, prove that

$$\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} abc & a^2 & a^3 \\ abc & b^2 & b^3 \\ abc & c^2 & c^3 \end{vmatrix}$$

Multiplying R_1, R_2, R_3 by a, b, c

$$= \frac{abc}{abc} \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

taking abc common from C_1

$$\Delta = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

s.

$$\begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

Q8 Prove that

$$\begin{vmatrix} 1 & x & xy \\ 1 & y & yx \\ 1 & y & xy \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & y & y^2 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 1 & x & xy \\ 1 & y & yx \\ 1 & y & xy \end{vmatrix}$$

Q19 Prove that

$$\begin{vmatrix} a & b & c & d \\ -b & a & -d & c \\ -c & d & a & -b \\ -d & -c & b & a \end{vmatrix} = (a^2 + b^2 + c^2 + d^2)^2$$

Soln.

$$\text{Let } \Delta = \begin{vmatrix} a & b & c & d \\ -b & a & -d & c \\ -c & d & a & -b \\ -d & -c & b & a \end{vmatrix}$$

$$= \frac{1}{abcd} \begin{vmatrix} a^2 & b^2 & c^2 & d^2 \\ -ab & ab & -cd & cd \\ -ac & bd & ac & -bd \\ -ad & -bc & bc & ad \end{vmatrix} \quad \begin{array}{l} \text{Multiplying } C_1, C_2, C_3, C_4 \text{ by} \\ a, b, c, d \text{ resp.} \end{array}$$

$$= \frac{1}{abcd} \begin{vmatrix} a^2+b^2+c^2+d^2 & b^2 & c^2 & d^2 \\ 0 & ab & -cd & cd \\ 0 & bd & ac & -bd \\ 0 & -bc & bc & ad \end{vmatrix} \quad C_1 + (C_2 + C_3 + C_4)$$

Expanding from C_1

$$= \frac{(a^2+b^2+c^2+d^2)}{abcd} \begin{vmatrix} ab & -cd & cd \\ bd & ac & -bd \\ -bc & bc & ad \end{vmatrix}$$

$$= \frac{(a^2+b^2+c^2+d^2)bcd}{abcd} \begin{vmatrix} a & -d & c \\ d & a & -b \\ -c & b & a \end{vmatrix}$$

taking b, c, d common from C_1, C_2, C_3

$$= \frac{(a^2+b^2+c^2+d^2)}{a} \begin{vmatrix} a & -d & c \\ d & a & -b \\ -c & b & a \end{vmatrix}$$

Expanding from R_1

$$\Delta = \frac{(a^2+b^2+c^2+d^2)}{a} \left\{ a(a^2+b^2) + d(ad-bc) + c(bd+ac) \right\}$$

$$= \frac{(a^2+b^2+c^2+d^2)}{a} \left\{ a^3 + ab^2 + ad^2 - \cancel{bd}d + \cancel{bd}d + ac^2 \right\}$$

$$= (a^2+b^2+c^2+d^2)(a^2+b^2+c^2+d^2)$$

$$\Delta = (a^2+b^2+c^2+d^2)^2$$

Q10. Prove that

$$(i) \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix}$$

$$= \begin{vmatrix} (b+c)^2 - a^2 & 0 & a^2 \\ 0 & (c+a)^2 - b^2 & b^2 \\ c^2 - (a+b)^2 & c^2 - (a+b)^2 & (a+b)^2 \end{vmatrix} \quad \begin{matrix} C_1 - C_3 \\ C_2 - C_3 \end{matrix}$$

$$= \begin{vmatrix} (b+c+a)(b+c-a) & 0 & a^2 \\ 0 & (c+a+b)(c+a-b) & b^2 \\ (c+a+b)(c-a-b) & (c+a+b)(c-a-b) & (a+b)^2 \end{vmatrix}$$

$$\Delta = (a+b+c)^2 \begin{vmatrix} (b+c-a) & 0 & a^2 \\ 0 & (c+a-b) & b^2 \\ (c-a-b) & (c-a-b) & (a+b)^2 \end{vmatrix} \quad \begin{array}{l} \text{taking } (a+b+c) \text{ Common} \\ \text{from } C_1 \text{ \& } C_2 \end{array}$$

$$= (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ -2b & -2a & 2ab \end{vmatrix} \quad R_3 - (R_1 + R_2)$$

$$= -2(a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ b & a & -ab \end{vmatrix}$$

Expanding from R_1

$$= -2(a+b+c)^2 \left\{ (b+c-a) \{ (-ab)(c+a-b) - ab^2 \} - 0 + a^2 \{ 0 - b(c+a-b) \} \right\}$$

$$= -2(a+b+c)^2 \left\{ (b+c-a) \{ -abc - a^2b + ab^2 - ab^2 \} + a^2 \{ -bc - ab + b^2 \} \right\}$$

$$= -2(a+b+c)^2 \left\{ (-ab)(b+c-a)(c+a) + (-ab)(ac + a^2 - ab) \right\}$$

$$= -2(a+b+c)^2 (-ab) \left\{ (b+c-a)(c+a) + ac + a^2 - ab \right\}$$

$$= 2ab(a+b+c)^2 \left\{ bc + ab + c^2 + ac - ac - a^2 + ac + a^2 - ab \right\}$$

$$= 2ab(a+b+c)^2 (bc + c^2 + ac)$$

$$= 2abc(a+b+c)^2 (b+c+a)$$

$$\Delta = 2abc(a+b+c)^3$$

(iii)

$$\begin{vmatrix} \frac{a^2+b^2}{c} & c & c \\ a & \frac{b^2+c^2}{a} & a \\ b & b & \frac{c^2+a^2}{b} \end{vmatrix} = 4abc$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} \frac{a^2+b^2}{c} & c & c \\ a & \frac{b^2+c^2}{a} & a \\ b & b & \frac{c^2+a^2}{b} \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} a^2+b^2 & c^2 & c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix} \quad \text{Multiplying } R_1, R_2, R_3 \text{ by } C, a, b \text{ resp.}$$

$$= \frac{1}{abc} \begin{vmatrix} a^2+b^2-c^2 & 0 & c^2 \\ 0 & b^2+c^2-a^2 & a^2 \\ b^2-c^2-a^2 & b^2-c^2-a^2 & c^2+a^2 \end{vmatrix} \quad \begin{matrix} C_1 - C_3 \\ C_2 - C_3 \end{matrix}$$

$$= \frac{1}{abc} \begin{vmatrix} a^2+b^2-c^2 & 0 & c^2 \\ 0 & b^2+c^2-a^2 & a^2 \\ -2a^2 & -2c^2 & 0 \end{vmatrix} \quad R_3 - (R_1 + R_2)$$

$$= \frac{-2}{abc} \begin{vmatrix} a^2+b^2-c^2 & 0 & c^2 \\ 0 & b^2+c^2-a^2 & a^2 \\ a^2 & c^2 & 0 \end{vmatrix} \quad \text{taking } -2 \text{ Common from } R_3$$

Expanding from R_1

$$= -\frac{2}{abc} \left\{ (a^2+b^2-c^2)(0-a^2c^2) - 0 + c^2(0-a^2(b^2+c^2-a^2)) \right\}$$

$$= -\frac{2}{abc} \left\{ -a^2c^2(a^2+b^2-c^2) - a^2c^2(b^2+c^2-a^2) \right\}$$

$$= -\frac{2}{abc}(-a^2c^2)\{a^2+b^2-c^2+c^2+b^2+a^2-a^2\}$$

$$= \frac{2ac}{b}(2b^2)$$

$$\Delta = 4abc$$

Q11 Prove that

$$\begin{vmatrix} x & a & a & a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} = (x+3a)(x-a)^3$$

Sol

$$\text{Let } \Delta = \begin{vmatrix} x & a & a & a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix}$$

$$= \begin{vmatrix} x+3a & a & a & a \\ x+3a & x & a & a \\ x+3a & a & x & a \\ x+3a & a & a & x \end{vmatrix} \quad C_1 + (C_2 + C_3 + C_4)$$

$$= (x+3a) \begin{vmatrix} 1 & a & a & a \\ 1 & x & a & a \\ 1 & a & x & a \\ 1 & a & a & x \end{vmatrix}$$

taking $(x+3a)$ Common from C_1

$$= (x+3a) \begin{vmatrix} 1 & a & a & a \\ 0 & x-a & 0 & 0 \\ 0 & 0 & x-a & 0 \\ 0 & 0 & 0 & x-a \end{vmatrix}$$

$$\begin{aligned} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{aligned}$$

$$\Delta = (x+3a)(x-a)^3 \begin{vmatrix} 1 & a & a & a \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \quad \begin{array}{l} \text{taking } (x-a) \text{ Common from} \\ R_2, R_3, R_4 \end{array}$$

Expanding from C_1

$$= (x+3a)(x-a)^3 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\Delta = (x+3a)(x-a)^3$$

Q12 Show that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ pa & qb & rc \end{vmatrix} = (a-b)(b-c)(c-a)$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ pa & qb & rc \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 1 \\ a-b & b-c & c \\ p(b-c) & q(c-a) & r(a-b) \end{vmatrix} \quad \begin{array}{l} C_1 - C_3 \\ C_2 - C_3 \end{array}$$

Expanding from R_1

$$= \begin{vmatrix} a-b & b-c \\ p(b-c) & q(c-a) \end{vmatrix}$$

$$= (a-b)(b-c) \begin{vmatrix} 1 & 1 \\ -p & -q \end{vmatrix} \quad \begin{array}{l} \text{taking } a-b, b-c \text{ Common} \\ \text{from } C_1, C_2 \text{ resp.} \end{array}$$

$$\Delta = (a-y)(b-y)(-a+b)$$

$$\Delta = (a-b)(b-y)(y-a)$$

Q13 Show that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (b-c)(c-a)(a-b)(a+b+c)$$

Sol:

$$\text{Let } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 1 \\ a-b & b-c & c \\ a^3-b^3 & b^3-c^3 & c^3 \end{vmatrix} \begin{matrix} C_1 - C_3 \\ C_2 - C_3 \end{matrix}$$

$$= \begin{vmatrix} 0 & 0 & 1 \\ (a-b) & (b-c) & c \\ (a-b)(a^2+ab+b^2) & (b-c)(b^2+bc+c^2) & c^3 \end{vmatrix}$$

$$= (a-b)(b-c) \begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & c \\ a^2+ab+b^2 & b^2+bc+c^2 & c^3 \end{vmatrix} \begin{matrix} \text{taking } (a-b), (b-c) \text{ Common} \\ \text{from } C_1 \text{ \& } C_2 \text{ resp.} \end{matrix}$$

Expanding from R_1

$$= (a-b)(b-c) \begin{vmatrix} 1 & 1 \\ a^2+ab+b^2 & b^2+bc+c^2 \end{vmatrix}$$

$$= (a-b)(b-c) \{ b^2+bc+c^2 - a^2-ab-b^2 \}$$

$$= (a-b)(b-c) \{ b^2 - a^2 + bc - ab \}$$

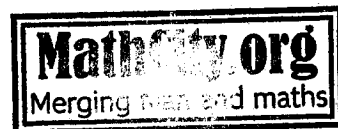
$$\Delta = (d-y)(p-y)\{(p-d)(p+d)+y(p-d)\}$$

$$= (d-y)(p-y)(p-d)(p+d+y)$$

$$\Delta = (p-y)(y-d)(d-p)(d+p+y)$$

Q14 Show that

$$\begin{vmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix} = (a+3)(a-1)^3$$



Soln-

$$\text{Let } \Delta = \begin{vmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix}$$

$$= \begin{vmatrix} a+3 & 1 & 1 & 1 \\ a+3 & a & 1 & 1 \\ a+3 & 1 & a & 1 \\ a+3 & 1 & 1 & a \end{vmatrix} \quad C_1 + (C_2 + C_3 + C_4)$$

$$= (a+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix} \quad \text{taking } a+3 \text{ Common from } C_1$$

$$= (a+3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & a-1 & 0 & 0 \\ 0 & 0 & a-1 & 0 \\ 0 & 0 & 0 & a-1 \end{vmatrix} \quad \begin{array}{l} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{array}$$

$$= (a+3) \begin{vmatrix} a-1 & 0 & 0 \\ 0 & a-1 & 0 \\ 0 & 0 & a-1 \end{vmatrix} \quad \text{Expanding from } C_1$$

$$\Delta = (a+3)(a-1)^3 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \quad \text{taking } a-1 \text{ Common from } R_1, R_2, R_3$$

$$\Delta = (a+3)(a-1)^3$$

$$(ii) \begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix} = abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right)$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix}$$

$$= abcd \begin{vmatrix} 1+\frac{1}{a} & \frac{1}{a} & \frac{1}{a} & \frac{1}{a} \\ \frac{1}{b} & 1+\frac{1}{b} & \frac{1}{b} & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & 1+\frac{1}{c} & \frac{1}{c} \\ \frac{1}{d} & \frac{1}{d} & \frac{1}{d} & 1+\frac{1}{d} \end{vmatrix} \quad \begin{array}{l} \text{taking } a, b, c, d \text{ Common from} \\ R_1, R_2, R_3, R_4 \text{ resp.} \end{array}$$

$$= abcd \begin{vmatrix} 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d} \\ \frac{1}{b} & 1+\frac{1}{b} & \frac{1}{b} & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & 1+\frac{1}{c} & \frac{1}{c} \\ \frac{1}{d} & \frac{1}{d} & \frac{1}{d} & 1+\frac{1}{d} \end{vmatrix} \quad \begin{array}{l} \text{Adding } R_2, R_3, R_4 \text{ in } R_1 \end{array}$$

$$= (abcd) \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) \begin{vmatrix} 1 & 1 & 1 & 1 \\ \frac{1}{b} & 1+\frac{1}{b} & \frac{1}{b} & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & 1+\frac{1}{c} & \frac{1}{c} \\ \frac{1}{d} & \frac{1}{d} & \frac{1}{d} & 1+\frac{1}{d} \end{vmatrix} \quad \begin{array}{l} \text{taking } (1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}) \\ \text{Common from } R_1 \end{array}$$

$$\Delta = abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) \begin{vmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{b} & 1 & 0 & 0 \\ \frac{1}{c} & 0 & 1 & 0 \\ \frac{1}{d} & 0 & 0 & 1 \end{vmatrix} \begin{matrix} C_2 - C_1 \\ C_3 - C_1 \\ C_4 - C_1 \end{matrix} \quad 461^{th}$$

$$= abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \quad \text{Expanding from } R_1$$

$$\Delta = abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right)$$



(iii)

$$\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} = x^2 y^2$$

Sol.

Let $\Delta = \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix}$

$$= \begin{vmatrix} 0 & 0 & 0 & 1 \\ -x & -x & 0 & 1 \\ -x & 0 & y & 1 \\ (-x+y+xy) & y & y & 1-y \end{vmatrix} \begin{matrix} C_1 - (1+x)C_4 \\ C_2 - C_4 \\ C_3 - C_4 \end{matrix}$$

Expanding from R_1

$$= - \begin{vmatrix} -x & -x & 0 \\ -x & 0 & y \\ -x+y+xy & y & y \end{vmatrix}$$

$$= - \begin{vmatrix} 0 & -x & 0 \\ -x & 0 & y \\ -x+xy & y & y \end{vmatrix}$$

$$= x \begin{vmatrix} 0 & 1 & 0 \\ -x & 0 & y \\ -x+xy & y & y \end{vmatrix}$$

taking $-x$ common from R_1

Expanding from R_1

$$= -x \begin{vmatrix} -x & y \\ -x+xy & y \end{vmatrix}$$

$$= -x(-xy + xy - xy^2)$$

$$= -x(-xy^2)$$

$$\Delta = x^2 y^2$$

Q15

$$\begin{vmatrix} a^3 & 3a^2 & 3a & 1 \\ a^2 & a^2+2a & 2a+1 & 1 \\ a & 2a+1 & a+2 & 1 \\ 1 & 3 & 3 & 1 \end{vmatrix}$$

Sol.

$$\text{let } \Delta = \begin{vmatrix} a^3 & 3a^2 & 3a & 1 \\ a^2 & a^2+2a & 2a+1 & 1 \\ a & 2a+1 & a+2 & 1 \\ 1 & 3 & 3 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a^3-1 & 3(a^2-1) & 3(a-1) & 0 \\ a^2-1 & a^2+2a-3 & 2(a-1) & 0 \\ a-1 & 2(a-1) & a-1 & 0 \\ 1 & 3 & 3 & 1 \end{vmatrix}$$

$$R_1 - R_4$$

$$R_2 - R_4$$

$$R_3 - R_4$$

Expanding from C_4

$$\Delta = \begin{vmatrix} a^3-1 & 3(a^2-1) & 3(a-1) \\ a^2-1 & a^2+2a-3 & 2(a-1) \\ a-1 & 2(a-1) & a-1 \end{vmatrix}$$

$$= \begin{vmatrix} (a-1)(a^2+a+1) & 3(a-1)(a+1) & 3(a-1) \\ (a-1)(a+1) & (a+3)(a-1) & 2(a-1) \\ a-1 & 2(a-1) & a-1 \end{vmatrix}$$

$$= (a-1)^3 \begin{vmatrix} a^2+a+1 & 3a+3 & 3 \\ a+1 & a+3 & 2 \\ 1 & 2 & 1 \end{vmatrix}$$

taking $a-1$ Common from C_1, C_2, C_3

$$= (a-1)^3 \begin{vmatrix} a^2+a-2 & 3a-3 & 0 \\ a-1 & a-1 & 0 \\ 1 & 2 & 1 \end{vmatrix}$$

$C_1 - 3C_3$
 $C_2 - 2C_3$

Expanding from C_3

$$= (a-1)^3 \begin{vmatrix} (a+2)(a-1) & 3(a-1) \\ a-1 & a-1 \end{vmatrix}$$

$$= (a-1)^5 \begin{vmatrix} a+2 & 3 \\ 1 & 1 \end{vmatrix}$$

taking $a-1$ Common from C_1 & C_2

$$= (a-1)^5 (a+2-3)$$

$$= (a-1)(a-1)$$

$$\Delta = (a-1)^6 \quad \text{--- Ans.}$$

Q16 If

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

44

if $A, B, C, \dots$ are Cofactors of $a, b, c, \dots$ in Δ ,
then show that

$$(i) \quad BC - F^2 = a\Delta$$

$$(ii) \quad CA - G^2 = b\Delta$$

$$(iii) \quad AB - H^2 = c\Delta$$

$$(iv) \quad GH - AF = f\Delta$$

$$(v) \quad HF - BG = g\Delta$$

$$(vi) \quad FG - CH = h\Delta$$

$$(vii) \quad aG + hF + gC = 0$$

$$(viii) \quad hG + bF + fC = 0$$

$$(ix) \quad gG + fF + cC = \Delta$$

Sol.

$$\text{Here } \Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

Expanding from R_1 ,

$$= a(bc - f^2) - h(ch - gf) + g(hf - gb)$$

$$= abc - af^2 - ch^2 + ghf + ghf - g^2b$$

$$\Delta = abc + 2ghf - af^2 - g^2b - ch^2 \quad \text{--- (1)}$$

Now

$$A = (-1)^{1+1} \begin{vmatrix} b & f \\ f & c \end{vmatrix} = (-1)^2 (bc - f^2) = bc - f^2$$

$$B = (-1)^{2+2} \begin{vmatrix} a & g \\ g & c \end{vmatrix} = (-1)^4 (ac - g^2) = ac - g^2$$

$$C = (-1)^{3+3} \begin{vmatrix} a & h \\ h & b \end{vmatrix} = (-1)^6 (ab - h^2) = ab - h^2$$

$$F = (-1)^{2+3} \begin{vmatrix} a & h \\ g & f \end{vmatrix} = (-1)^5 (af - gh) = gh - af$$

$$G = (-1)^{1+3} \begin{vmatrix} h & b \\ g & f \end{vmatrix} = (-1)^4 (hf - gb) = hf - gb$$

45

$$H = (-1)^{1+2} \begin{vmatrix} h & f \\ g & c \end{vmatrix} = (-1)^3 (ch - gf) = gf - ch$$

Now

$$\begin{aligned} \text{(i)} \quad BC - F^2 &= (ac - g^2)(ab - h^2) - (gh - af)^2 \\ &= abc - ach^2 - abg^2 + g^2h^2 - g^2h^2 - a^2f^2 + 2afgh \\ &= abc - ach^2 - abg^2 - a^2f^2 + 2afgh \\ &= a(abc + 2fgh - af^2 - g^2b - ch^2) \end{aligned}$$

$$\text{So } BC - F^2 = a\Delta$$

$$\text{(ii)} \quad CA - G^2 = b\Delta$$

Sol.

$$\begin{aligned} CA - G^2 &= (ab - h^2)(bc - f^2) - (hf - gb)^2 \\ &= abc - abf^2 - h^2bc + h^2f^2 - h^2f^2 - g^2b^2 + 2fghb \\ &= b(abc + 2fgh - af^2 - g^2b - ch^2) \end{aligned}$$

$$CA - G^2 = b\Delta$$

$$\text{(iii)} \quad AB - H^2 = c\Delta$$

Sol.

$$\begin{aligned} AB - H^2 &= (bc - f^2)(ac - g^2) - (gf - hc)^2 \\ &= abc^2 - g^2bc - f^2ac + g^2f^2 - g^2f^2 - c^2h^2 + 2ghfc \\ &= c(abc + 2fgh - af^2 - g^2b - ch^2) \end{aligned}$$

$$AB - H^2 = c\Delta$$

$$\text{(iv)} \quad GH - AF = f\Delta$$

Sol.

$$\begin{aligned} GH - AF &= (hf - gb)(fg - ch) - (bc - f^2)(gh - af) \\ &= f^2gh - ch^2f - bfg^2 + bcfh - bcfh + abcf + f^2gh - af^3 \end{aligned}$$

$$\begin{aligned} GH - AF &= 2f^2gh - ch^2f - bfg^2 + abcf - af^3 \\ &= f(abc + 2fgh - af^2 - g^2b - ch^2) \end{aligned}$$

$$GH - AF = f\Delta$$

$$(v) \quad HF - BG = g\Delta$$

Sol.

$$\begin{aligned} HF - BG &= (fg - ch)(gh - af) - (ac - g^2)(hf - gb) \\ &= g^2fh - af^2g - gch^2 + afgh - afgh + acgb + g^2hf - g^3b \\ &= abcg + 2g^2hf - af^2g - gch^2 - g^3b \\ &= g(abc + 2ghf - af^2 - ch^2 - g^2b) \end{aligned}$$

$$HF - BG = g\Delta$$

$$(vi) \quad FG - CH = h\Delta$$

Sol.

$$\begin{aligned} FG - CH &= (gh - af)(hf - gb) - (ab - h^2)(gf - ch) \\ &= gh^2f - g^2hb - ahf^2 + afgf - afgf + abch + h^2gf - ch^2 \\ &= abch + 2gh^2f - ahf^2 - g^2hb - ch^2 \\ &= h(abc + 2ghf - af^2 - g^2b - ch^2) \end{aligned}$$

$$FG - CH = h\Delta$$

$$(vii) \quad aG + hF + gC = 0$$

Sol.

$$\begin{aligned} aG + hF + gC &= a(hf - bg) + h(gh - af) + g(ab - h^2) \\ &= ahf - abg + gh^2 - ahf + gab - gh^2 \\ &= 0 \end{aligned}$$

$$(viii) \quad hG + bF + fC = 0$$

Sol.

$$\begin{aligned}
 hG + bF + fC &= h(hf - bg) + b(gh - af) + f(ab - h^2) \\
 &= hf - bg h + bgh - af^2 + af^2 - h^2 f \\
 &= 0
 \end{aligned}$$

$$(ix) \quad gG + fF + cC = \Delta$$

Sol.

$$\begin{aligned}
 gG + fF + cC &= g(hf - bg) + f(gh - af) + c(ab - h^2) \\
 &= ghf - g^2b + ghf - af^2 + abc - ch^2 \\
 &= abc + 2ghf - af^2 - g^2b - ch^2
 \end{aligned}$$

$$gG + fF + cC = \Delta$$

Q17 If Δ of problem 16 is zero, show that

$$(i) \quad F^2 + G^2 = C(A+B)$$

$$(ii) \quad G^2 + H^2 = A(B+C)$$

$$(iii) \quad H^2 + F^2 = B(A+C)$$

$$(iv) \quad ABC = FGH$$

Sol.

(i) As we have proved that

$$\left. \begin{aligned} BC - F^2 &= a\Delta \\ + \quad CA - G^2 &= b\Delta \end{aligned} \right\}$$

$$\text{But } \Delta = 0$$

So

$$\left. \begin{aligned} BC - F^2 &= 0 \\ CA - G^2 &= 0 \end{aligned} \right\}$$

$$\left. \begin{aligned} F^2 &= BC \\ G^2 &= CA \end{aligned} \right\}$$

$$\begin{aligned} F^2 + G^2 &= BC + CA \\ F^2 + G^2 &= C(B+A) \end{aligned}$$

$$\text{or } F^2 + G^2 = C(A+B)$$

$$(ii) \quad G^2 + H^2 = A(B+C)$$

Sol. As we have proved that

$$\left. \begin{aligned} CA - G^2 &= b\Delta \\ + AB - H^2 &= c\Delta \end{aligned} \right\}$$

$$\text{But } \Delta = 0$$

S.

$$\left. \begin{aligned} CA - G^2 &= 0 \\ AB - H^2 &= 0 \end{aligned} \right\}$$

or

$$\left. \begin{aligned} G^2 &= CA \\ H^2 &= AB \end{aligned} \right\}$$

Adding

$$\begin{aligned} G^2 + H^2 &= CA + AB \\ &= A(C+B) \\ G^2 + H^2 &= A(B+C) \end{aligned}$$

$$(iii) \quad H^2 + F^2 = B(A+C)$$

Sol. As we have proved that

$$\left. \begin{aligned} AB - H^2 &= c\Delta \\ + BC - F^2 &= a\Delta \end{aligned} \right\}$$

$$\text{But } \Delta = 0$$

S.

$$\left. \begin{aligned} AB - H^2 &= 0 \\ BC - F^2 &= 0 \end{aligned} \right\}$$

$$\left. \begin{aligned} H^2 &= AB \\ + F^2 &= BC \end{aligned} \right\}$$

Adding

$$H^2 + F^2 = B(A+C)$$

(iv) $ABC = FGH$

Sol. As we know that

$$\left. \begin{aligned} BC &= F^2 \\ CA &= G^2 \\ AB &= H^2 \end{aligned} \right\}$$

Multiplying these eqs.

$$(BC)(CA)(AB) = F^2 G^2 H^2$$

$$A^2 B^2 C^2 = F^2 G^2 H^2$$

$$\therefore ABC = FGH$$

Available at
www.mathcity.org

Q18 Prove that

$$\begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix}^2 = \begin{vmatrix} b^2+c^2 & ab & ac \\ ba & c^2+a^2 & bc \\ ca & cb & a^2+b^2 \end{vmatrix}$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix} \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 0+c^2+b^2 & 0+0+ab & 0+ac+0 \\ 0+0+ab & c^2+0+a^2 & bc+0+0 \\ 0+ac+0 & bc+0+0 & b^2+a^2+0 \end{vmatrix}$$

$$= \begin{vmatrix} b^2+c^2 & ab & ac \\ ba & c^2+a^2 & bc \\ ca & cb & a^2+b^2 \end{vmatrix}$$

Q19 Show that

$$\begin{vmatrix} 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega \\ 1 & \omega^3 & \omega & \omega^4 \\ 1 & \omega^4 & \omega^3 & \omega^2 \end{vmatrix}^2 = 125$$

where ω is a fifth root of 1.

Sol.

Since ω is the fifth root of 1

$$\text{So } \omega^5 = 1 \quad \text{--- (1)}$$

$$\text{+ } 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0 \quad \text{--- (2)}$$

Now

$$\begin{aligned} \text{let } \Delta &= \begin{vmatrix} 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega \\ 1 & \omega^3 & \omega & \omega^4 \\ 1 & \omega^4 & \omega^3 & \omega^2 \end{vmatrix}^2 \\ &= \begin{vmatrix} 1 & \omega & \omega^2 & \omega^3 & 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega & 1 & \omega^2 & \omega^4 & \omega \\ 1 & \omega^3 & \omega & \omega^4 & 1 & \omega^3 & \omega & \omega^4 \\ 1 & \omega^4 & \omega^3 & \omega^2 & 1 & \omega^4 & \omega^3 & \omega^2 \end{vmatrix} \\ &= \begin{vmatrix} 1+\omega+\omega^2+\omega^3 & \omega+\omega^3+\omega^5+\omega^7 & \omega^2+\omega^5+\omega^7+\omega^6 & \omega^3+\omega^2+\omega^6+\omega^5 \\ 1+\omega^2+\omega^4+\omega & \omega+\omega^4+\omega^7+\omega^5 & \omega^2+\omega^6+\omega^5+\omega^4 & \omega^3+\omega^3+\omega^8+\omega^3 \\ 1+\omega^3+\omega+\omega^4 & \omega+\omega^5+\omega^4+\omega^8 & \omega^2+\omega^7+\omega^2+\omega^7 & \omega^3+\omega^4+\omega^5+\omega^6 \\ 1+\omega^4+\omega^3+\omega^2 & \omega+\omega^6+\omega^6+\omega^6 & \omega^2+\omega^8+\omega^4+\omega^5 & \omega^3+\omega^5+\omega^7+\omega^4 \end{vmatrix} \\ &= \begin{vmatrix} 1+\omega+\omega^3+\omega^3 & 1+\omega+\omega^2+\omega^3 & 1+\omega+\omega^2+\omega^3 & 1+\omega+\omega^3+\omega^3 \\ 1+\omega+\omega^3+\omega^4 & 1+\omega+\omega^2+\omega^4 & 1+\omega+\omega^2+\omega^4 & 4\omega^3 \\ 1+\omega+\omega^3+\omega^4 & 1+\omega+\omega^3+\omega^4 & 4\omega^2 & 1+\omega+\omega^3+\omega^4 \\ 1+\omega^2+\omega^3+\omega^4 & 4\omega & 1+\omega^2+\omega^3+\omega^4 & 1+\omega^2+\omega^3+\omega^4 \end{vmatrix} \quad \text{By (1)} \end{aligned}$$

$$\Delta = \begin{vmatrix} -\omega^4 & -\omega^4 & -\omega^4 & -\omega^4 \\ -\omega^3 & -\omega^3 & -\omega^3 & 4\omega^3 \\ -\omega^2 & -\omega^2 & 4\omega^2 & -\omega^2 \\ -\omega & 4\omega & -\omega & -\omega \end{vmatrix}$$

$$= (-1)^4 \omega^4 \cdot \omega^3 \cdot \omega^2 \cdot \omega \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -4 \\ 1 & 1 & -4 & 1 \\ 1 & -4 & 1 & 1 \end{vmatrix}$$

$$= \omega^8 \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & -5 \\ 0 & 0 & -5 & 0 \\ 0 & -5 & 0 & 0 \end{vmatrix}$$

$R_2 - R_1$
 $R_3 - R_1$
 $R_4 - R_1$

$$= (\omega^8)^2 \begin{vmatrix} 0 & 0 & -5 \\ 0 & -5 & 0 \\ -5 & 0 & 0 \end{vmatrix} \quad \text{expanding from } C_1$$

$$= (1)^2 \begin{vmatrix} 0 & 0 & -5 \\ 0 & -5 & 0 \\ -5 & 0 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & -5 \\ 0 & -5 & 0 \\ -5 & 0 & 0 \end{vmatrix}$$

Expanding from R_1

$$= 0 - 0 + (-5) \begin{vmatrix} 0 & -5 \\ -5 & 0 \end{vmatrix}$$

$$= -5(0 - 25)$$

$$\Delta = 125$$

Q20 Prove that the determinant

$$\begin{vmatrix} 2b_1 + c_1 & c_1 + 3a_1 & 2a_1 + 3b_1 \\ 2b_2 + c_2 & c_2 + 3a_2 & 2a_2 + 3b_2 \\ 2b_3 + c_3 & c_3 + 3a_3 & 2a_3 + 3b_3 \end{vmatrix}$$

is a multiple of the determinant

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

and find the other factor.

Sol.

Consider

$$\begin{vmatrix} 2b_1 + c_1 & c_1 + 3a_1 & 2a_1 + 3b_1 \\ 2b_2 + c_2 & c_2 + 3a_2 & 2a_2 + 3b_2 \\ 2b_3 + c_3 & c_3 + 3a_3 & 2a_3 + 3b_3 \end{vmatrix}$$

$$= \begin{vmatrix} 0a_1 + 2b_1 + 1 \cdot c_1 & 3a_1 + 0b_1 + 1 \cdot c_1 & 2a_1 + 3b_1 + 0c_1 \\ 0a_2 + 2b_2 + 1 \cdot c_2 & 3a_2 + 0b_2 + 1 \cdot c_2 & 2a_2 + 3b_2 + 0c_2 \\ 0a_3 + 2b_3 + 1 \cdot c_3 & 3a_3 + 0b_3 + 1 \cdot c_3 & 2a_3 + 3b_3 + 0c_3 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \begin{vmatrix} 0 & 3 & 2 \\ 2 & 0 & 3 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \cdot \{ 0 - 3(0 - 3) + 2(2 - 0) \}$$

$$= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} (9 + 4)$$

$$= 13 \cdot \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Hence given determinant is a multiple of $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

& other factor is 13.

Q21 35,282 ; 44,759 ; 58,916 ; 80,652 & 92,469 are all multiples of 13. Show that the determinant

$$\begin{vmatrix} 3 & 5 & 2 & 8 & 2 \\ 4 & 4 & 7 & 5 & 9 \\ 5 & 8 & 9 & 1 & 6 \\ 8 & 0 & 6 & 5 & 2 \\ 9 & 2 & 4 & 6 & 9 \end{vmatrix} \text{ is also a multiple of 13.}$$

Sol.

Let $\Delta = \begin{vmatrix} 3 & 5 & 2 & 8 & 2 \\ 4 & 4 & 7 & 5 & 9 \\ 5 & 8 & 9 & 1 & 6 \\ 8 & 0 & 6 & 5 & 2 \\ 9 & 2 & 4 & 6 & 9 \end{vmatrix}$

$$= \begin{vmatrix} 3 & 5 & 2 & 8 & 35,282 \\ 4 & 4 & 7 & 5 & 44,759 \\ 5 & 8 & 9 & 1 & 58,916 \\ 8 & 0 & 6 & 5 & 80,652 \\ 9 & 2 & 4 & 6 & 92,469 \end{vmatrix}$$

$$R_5 + 10R_4 + 100R_3 + 1000R_2 + 10000R_1$$

Since 35,282, 44,759, 58,916, 80,652 & 92,469 are all multiples of 13, so 13 is a common factor of these numbers.

Hence

$$\Delta = 13 \begin{vmatrix} 3 & 5 & 2 & 8 & 2714 \\ 4 & 4 & 7 & 5 & 2443 \\ 5 & 8 & 9 & 1 & 4532 \\ 8 & 0 & 6 & 5 & 6204 \\ 9 & 2 & 4 & 6 & 7113 \end{vmatrix}$$

taking 13 Common from C_5 So given determinant Δ is a multiple of 13.

Q22 Prove that

$$\begin{vmatrix} (a-x)^2 & (a-y)^2 & (a-z)^2 \\ (b-x)^2 & (b-y)^2 & (b-z)^2 \\ (c-x)^2 & (c-y)^2 & (c-z)^2 \end{vmatrix} = 2(a-b)(b-c)(c-a)(x-y)(y-z)(z-x)$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} (a-x)^2 & (a-y)^2 & (a-z)^2 \\ (b-x)^2 & (b-y)^2 & (b-z)^2 \\ (c-x)^2 & (c-y)^2 & (c-z)^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 - 2ax + x^2 & a^2 - 2ay + y^2 & a^2 - 2az + z^2 \\ b^2 - 2bx + x^2 & b^2 - 2by + y^2 & b^2 - 2bz + z^2 \\ c^2 - 2cx + x^2 & c^2 - 2cy + y^2 & c^2 - 2cz + z^2 \end{vmatrix}$$

$$= \begin{vmatrix} a^2 & -2a & 1 \\ b^2 & -2b & 1 \\ c^2 & -2c & 1 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$\Delta = \Delta_1 \cdot \Delta_2 \quad \text{--- (1)}$$

Now

$$\Delta_1 = \begin{vmatrix} a^2 & -2a & 1 \\ b^2 & -2b & 1 \\ c^2 & -2c & 1 \end{vmatrix}$$

$$= -2 \begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix}$$

$$= -2 \begin{vmatrix} a^2 - c^2 & a - c & 0 \\ b^2 - c^2 & b - c & 0 \\ c^2 & c & 1 \end{vmatrix}$$

 $R_1 - R_3$ $R_2 - R_3$ Expanding from C_3

$$= -2 \begin{vmatrix} a^2 - c^2 & a - c \\ b^2 - c^2 & b - c \end{vmatrix}$$

$$= -2(a-c)(b-c) \begin{vmatrix} a+c & 1 \\ b+c & 1 \end{vmatrix}$$

taking $a-c$ Common from R_1
 & $b-c$ Common from R_2

$$= -2(a-c)(b-c)(a+c-b-c)$$

$$\Delta_1 = 2(a-b)(b-c)(c-a)$$

$$\Delta_2 = \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 1 \\ x-z & y-z & z \\ x^2-z^2 & y^2-z^2 & z^2 \end{vmatrix}$$

 $C_1 - C_3$ $C_2 - C_3$ Expanding from R_1

$$= \begin{vmatrix} x-z & y-z \\ x^2-z^2 & y^2-z^2 \end{vmatrix}$$

$$\Delta_1 = (x-z)(y-z) \begin{vmatrix} 1 & 1 \\ x+z & y+z \end{vmatrix} \quad \text{taking } x-z, y-z \text{ Common from } R_2 \text{ \& } R_1.$$

$$= (x-z)(y-z)(y+z-x-z)$$

$$= (x-z)(y-z)(y-x)$$

$$\Delta_2 = (x-y)(y-z)(z-x)$$

Putting values of Δ_1, Δ_2 in ①

$$\Delta = 2(a-b)(b-c)(c-a)(x-y)(y-z)(z-x)$$

Q23 Show that

$$\begin{vmatrix} 2bc-a^2 & c^2 & b^2 \\ c^2 & 2ca-b^2 & a^2 \\ b^2 & a^2 & 2ab-c^2 \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix} = (a^3+b^3+c^3-3abc)$$

Sol.

$$\text{Let } \Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \cdot \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \cdot \begin{vmatrix} -a & -b & -c \\ c & a & b \\ b & c & a \end{vmatrix} \quad \text{taking transpose of second det.}$$

$$= \begin{vmatrix} -a^2+bc+bc & -ab+ab+c^2 & -ac+b^2+a/c \\ -ab+c^2+ab & -b^2+ca+ca & -bc+bc+a^2 \\ -ac+ac+b^2 & -bc+a^2+bc & -c^2+ab+ab \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 2bc-a^2 & c^2 & b^2 \\ c^2 & 2ca-b^2 & a^2 \\ b^2 & a^2 & 2ab-c^2 \end{vmatrix} \quad \text{--- Ans I}$$

Again Consider

$$\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix}$$

$$\Delta = \Delta_1 \cdot \Delta_2 \quad \text{--- (1)}$$

Now

$$\Delta_1 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

Expanding from R_1

$$= a(bc - a^2) - b(b^2 - ac) + c(ab - c^2)$$

$$= abc - a^3 - b^3 + abc + abc - c^3$$

$$= -a^3 - b^3 - c^3 + 3abc$$

$$\Delta_1 = -(a^3 + b^3 + c^3 - 3abc)$$

$$\Delta_2 = \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix}$$

Expanding from R_1

$$= (-a)(a^2 - bc) - c(-ab + c^2) + b(-b^2 + ac)$$

$$= -a^3 + abc + abc - c^3 - b^3 + abc$$

$$= -a^3 - b^3 - c^3 + 3abc$$

$$\Delta_2 = -(a^3 + b^3 + c^3 - 3abc)$$

Put in (1)

$$\Delta = (a^3 + b^3 + c^3 - 3abc)^2$$

Hence

$$\begin{vmatrix} 2bc - a^2 & c^2 & b^2 \\ c^2 & 2ca - b^2 & a^2 \\ b^2 & a^2 & 2ab - c^2 \end{vmatrix} = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \begin{vmatrix} -a & c & b \\ -b & a & c \\ -c & b & a \end{vmatrix} = (a^3 + b^3 + c^3 - 3abc)^2$$

Q.24 Find, by the adjoint method, the inverse of each⁵⁰ of the following matrices:

(i) $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

Sol.

Let $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$

We know that

$$A^{-1} = \frac{\text{Adj } A}{|A|} \quad \text{--- (1)}$$

Now

$$|A| = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix}$$

Expanding from R_1

$$= 2(4-1) - 1(2-1) + 1(1-2)$$

$$= 2(3) - (1) + (-1)$$

$$= 6 - 1 - 1$$

$$|A| = 4 \neq 0 \quad \text{So } A^{-1} \text{ exists}$$

Now

$$\text{Adj } A = \begin{bmatrix} \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} \\ -\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} \end{bmatrix}^t$$

$$= \begin{bmatrix} 4-1 & -(2-1) & 1-2 \\ -(2-1) & 4-1 & -(2-1) \\ 1-2 & -(2-1) & 4-1 \end{bmatrix}^t$$

$$= \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}^t$$

$$\text{Adj } A = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$$

So from ①

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$$

$$\text{So } A^{-1} = \begin{bmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

(ii) $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$

Sol:

$$\text{Let } D = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$$

We know that

$$D^{-1} = \frac{\text{Adj } D}{|D|} \quad \text{--- ①}$$

Now

$$|D| = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

Expanding from R_1 ,

$$= a(bc - f^2) - h(ch - gf) + g(hf - gb)$$

$$= abc - af^2 - ch^2 + ghf + ghf - g^2b$$

$$|D| = abc + 2ghf - af^2 - ch^2 - g^2b \neq 0 \text{ (say) then } D^{-1} \text{ exists.}$$

Now

$$\text{Adj } D = \begin{bmatrix} \begin{vmatrix} b & f \\ f & c \end{vmatrix} & -\begin{vmatrix} h & f \\ g & c \end{vmatrix} & \begin{vmatrix} h & b \\ g & f \end{vmatrix} \\ -\begin{vmatrix} h & g \\ f & c \end{vmatrix} & \begin{vmatrix} a & g \\ g & c \end{vmatrix} & -\begin{vmatrix} a & h \\ g & f \end{vmatrix} \\ \begin{vmatrix} h & g \\ b & f \end{vmatrix} & -\begin{vmatrix} a & g \\ h & f \end{vmatrix} & \begin{vmatrix} a & h \\ h & b \end{vmatrix} \end{bmatrix}^t$$

$$= \begin{bmatrix} bc - f^2 & -(ch - gf) & hf - gb \\ -(ch - gf) & ac - g^2 & af - gh \\ hf - gb & -(af - gh) & ab - h^2 \end{bmatrix}^t$$

$$= \begin{bmatrix} bc - f^2 & gf - ch & hf - gb \\ gf - ch & ac - g^2 & gh - af \\ hf - gb & gh - af & ab - h^2 \end{bmatrix}^t$$

$$\text{Adj } D = \begin{bmatrix} bc - f^2 & gf - ch & hf - gb \\ gf - ch & ac - g^2 & gh - af \\ hf - gb & gh - af & ab - h^2 \end{bmatrix}$$

So from ①

$$\bar{D}^{-1} = \frac{1}{(abc+eghf-af^2-ch^2-g^2b)} \begin{bmatrix} bc-f^2 & gf-ch & hf-gb \\ gf-ch & ac-g^2 & gh-af \\ hf-gb & gh-af & ab-h^2 \end{bmatrix}$$

(iii) $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & -1 \\ 3 & 2 & -2 \end{bmatrix}$

Sol.

Let $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & -1 \\ 3 & 2 & -2 \end{bmatrix}$

We know that

$$\bar{A}^{-1} = \frac{\text{Adj } A}{|A|} \quad \text{--- ①}$$

Now

$$|A| = \begin{vmatrix} 1 & -1 & 2 \\ 2 & 1 & -1 \\ 3 & 2 & -2 \end{vmatrix}$$

Expanding from R_1

$$= 1(-2+2) + 1(-4+3) + 2(4-3)$$

$$= 0 - 1 + 2$$

$$|A| = 1 \neq 0, \text{ so } \bar{A}^{-1} \text{ exists.}$$

Now

$$\text{Adj } A = \begin{bmatrix} \begin{vmatrix} 1 & -1 \\ 2 & -2 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} \\ -\begin{vmatrix} -1 & 2 \\ 2 & -2 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix} & -\begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix} \\ \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} \end{bmatrix}^t$$

$$\text{Adj } A = \begin{bmatrix} -2+2 & -(-4+3) & 4-3 \\ -(2-4) & -2-6 & -(2+3) \\ 1-2 & -(-1-4) & 1+2 \end{bmatrix}^t$$

$$= \begin{bmatrix} 0 & 1 & 1 \\ 2 & -8 & -5 \\ -1 & 5 & 3 \end{bmatrix}^t$$

$$\text{Adj } A = \begin{bmatrix} 0 & 2 & -1 \\ 1 & -8 & 5 \\ 1 & -5 & 3 \end{bmatrix}$$

So from ①

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 0 & 2 & -1 \\ 1 & -8 & 5 \\ 1 & -5 & 3 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 0 & 2 & -1 \\ 1 & -8 & 5 \\ 1 & -5 & 3 \end{bmatrix}$$